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# AI-Assisted Licensing Document Development

## An ISU AGN-201M / NUREG-1537 Test Case

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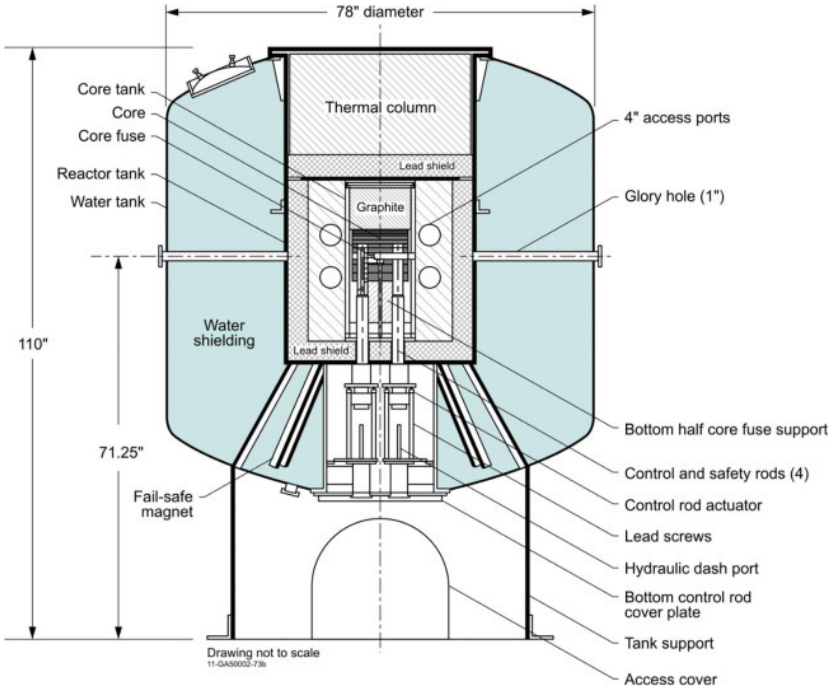
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Battelle Energy Alliance manages INL for the
U.S. Department of Energy's Office of Nuclear Energy

Idaho National Laboratory

# Motivation & Problem Statement

- **Use case:** ISU test reactor (AGN-201M)
  - Need: Draft license renewal to be submitted to the NRC
  - Guidance provided by the NRC NUREG-1537 document
    - Part 1: Section by section instructions
    - Part 2: Reviewer guidance
- **Licensing is a data-management problem as much as a writing one**
- An FSAR must be
  - **Traceable:** Every claim tied to a source document
  - **Consistent:** The same parameter agrees across chapters
  - **Complete:** Every required topic addressed or dispositioned
- **The corpus:** Hundreds of documents across PDF, DOCX, XLSX, superseded and current versions
  - Decades of records → difficult evidence reconciliation → traceability / consistency / completeness issues → Request for additional information and schedule burden



# Why Naive AI Fails and Proposed Approach

- **The naive approach:** Ask an LLM to “write the section”
  - Result: Fluent prose, but claims with no source, invented citations, and no signal for what's missing
- **The key distinction**

**Evidence supply**
Retrieval from the applicant’s own controlled documents

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**Evidence articulation**
LLM only organizes and phrases what was retrieved

- **Risks**
  - Commission: Ungrounded text enters the FSAR
  - Omission: A topic that could be drafted is left blank
- **Goal:** An AI-assisted workflow for drafting FSAR sections according to licensing directives (e.g., based on NUREG-1537)
  - Applicant evidence → controlled retrieval → LLM articulation → human review
  - LLM is an articulate assistant, not an author of record (i.e., it does not supply the facts)
  - Every retained claim must trace to a citable chunk, or be flagged as a gap

- ✗ No citations
- ✗ Fabricated references
- ✗ Silent omissions
- ✗ Wrong facts stated fluently

- ✓ Every claim cited to a source
- ✓ Missing topic declared (not hidden)
- ✓ Fact bounded by the evidence
- ✓ Human reviewer stays accountable

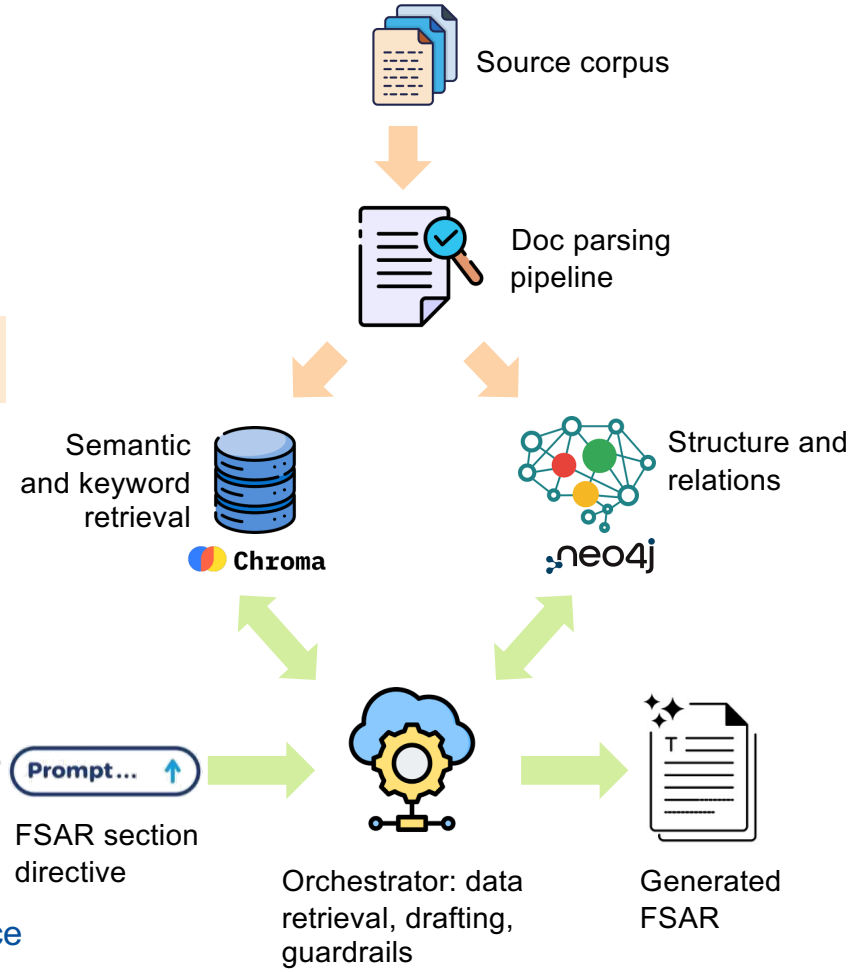
The reactor is licensed for a steady-state power of 5 kW, cooled by natural air convection.
**(Source: Chunk 12)**

The control and safety instrumentation is analog solid-state.
**(Source: Chunk 27)**

**[INFORMATION NOT AVAILABLE: control-rod technical specifications]**

# Architecture + Guardrails

- **Design principle:** Retrieval-first, evidence-driven drafting
  - No free-form generation without sources
- Two pipelines
  - **Prepare evidence:** Parse heterogeneous documents into cited chunks, annotate entities/parameters, populate stores
  - **Draft & review:** Retrieve, assemble evidence package, generate cited text, run guardrails
    - Layer 1: Every claim keyed to a chunk
    - Layer 2: Identify missing topics
    - Layer 3: Flags weakly supported sentences
    - LLM verification pass (optional)
- **Knowledge graph:** System Model-based System Engineering (MBSE) model provides component and functional context
  - Links evidence to the components, parameters, and requirements it describes
  - Retrieval finds facility-specific evidence, not generic guidance
- **Local-first:** Models and stores ran locally
  - Applicant material never left the environment



# Current Status: Results

- **What works**
  - 53 subsections configured, 45 drafted, 8 correctly declared not-applicable
  - KG-enabled retrieval reduced unsupported technical claims in the full-run comparison
  - Drafting reached 91% and 100% factual-consistency (§4.2.2 and §7.3)
- **What the workflow exposed**
  - Missing evidence
  - Incorrect applicability
  - Weakly grounded claims
  - Tradeoffs between retrieval depth and runtime

- **What needs improvement**
  - Coverage is partial: §4.2.2 and §7.3 land around 71–75% topic coverage
    - Remaining gaps reflect corpus completeness and directive tuning
  - Unit normalization (e.g., “500 psi” vs. “3.45 MPa”) is flagged as false conflict
  - Recursive-RAG tradeoff: It reduces gap markers but costs ~4× generation time
  - Full MBSE integration and broader validation remain future work

# Path Forward

"*We started with a document-drafting problem and discovered an even bigger opportunity: managing the engineering evidence behind the corpus*"

- The bigger opportunity: Drafting text-from-text is step one
  - The real goal is a structured **engineering-evidence platform**
  - Engineering artifacts (e.g., requirements, system models, FMEAs, PRA analysis, and simulation results) become citable evidence with the same provenance discipline

